

Definition. Let G be a region and let $a = a + bi = (a, b) \in G$. A function $g : G \rightarrow \mathbb{R}$ is differentiable at a if there exist constants A and B and functions $e_1, e_2 : G \rightarrow \mathbb{R}$ which are continuous at a with $e_1(a, b) = e_2(a, b) = 0$ such that

$$g(x, y) = g(a, b) + (x - a)\{A + e_1(x, y)\} + (y - b)\{B + e_2(x, y)\}$$

If g is differentiable at a , we will define $g_x(a, b) = A$ and $g_y(a, b) = B$.

Proposition. Let G be a region and let $a = a + bi = (a, b) \in G$. Suppose that $u, v : G \rightarrow \mathbb{R}$ such that u, v are (continuously) differentiable at a and that u, v satisfy the Cauchy-Riemann equations at a . Then, $f = u + iv$ is (continuously) differentiable at a .

Proof. Since u, v are differentiable at a , there exist functions $e_r : G \rightarrow \mathbb{R}$, $r = 1, 2, 3, 4$, which are continuous at a with $e_r(a, b) = 0$, $r = 1, 2, 3, 4$, such that

$$u(x, y) = u(a, b) + (x - a)\{u_x(a, b) + e_1(x, y)\} + (y - b)\{u_y(a, b) + e_2(x, y)\} \quad (1)$$

$$v(x, y) = v(a, b) + (x - a)\{v_x(a, b) + e_3(x, y)\} + (y - b)\{v_y(a, b) + e_4(x, y)\} \quad (2)$$

If in (1) we replace $u_y(a, b) = -v_x(a, b) = i \cdot i \cdot v_x(a, b)$ and in (2) we replace $v_y(a, b) = u_x(a, b)$ and then multiply (2) by i and add the result to (1) we obtain (after a rearrangement)

$$f(z) - f(a) = (z - a)[u_x(a, b) + i v_x(a, b) + w(z)]$$

where

$$w(z) = \frac{x - a}{z - a}\{e_1(x, y) + ie_3(x, y)\} + \frac{y - b}{z - a}\{e_2(x, y) + ie_4(x, y)\}.$$

Since $|w(z)| \leq |e_1(x, y)| + |e_2(x, y)| + |e_3(x, y)| + |e_4(x, y)|$, then

$$\lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a} = \lim_{z \rightarrow a} u_x(a, b) + i v_x(a, b) + w(z) = u_x(a, b) + i v_x(a, b)$$

since $\lim_{z \rightarrow a} w(z) = 0$.